

HOUSE No. 3539**The Commonwealth of Massachusetts**

PRESENTED BY:

Steven Owens and Jennifer Balinsky Armini*To the Honorable Senate and House of Representatives of the Commonwealth of Massachusetts in General Court assembled:*

The undersigned legislators and/or citizens respectfully petition for the adoption of the accompanying bill:

An Act relative to a tactical transition to affordable, clean thermal energy.

PETITION OF:

NAME:	DISTRICT/ADDRESS:	DATE ADDED:
<i>Steven Owens</i>	<i>29th Middlesex</i>	<i>1/17/2025</i>
<i>Samantha Montaño</i>	<i>15th Suffolk</i>	<i>3/4/2025</i>
<i>Greg Schwartz</i>	<i>12th Middlesex</i>	<i>3/5/2025</i>
<i>Lindsay N. Sabadosa</i>	<i>1st Hampshire</i>	<i>3/5/2025</i>
<i>Kenneth I. Gordon</i>	<i>21st Middlesex</i>	<i>3/19/2025</i>
<i>Carmine Lawrence Gentile</i>	<i>13th Middlesex</i>	<i>4/4/2025</i>
<i>Mike Connolly</i>	<i>26th Middlesex</i>	<i>4/4/2025</i>
<i>James C. Arena-DeRosa</i>	<i>8th Middlesex</i>	<i>4/5/2025</i>
<i>Tommy Vitolo</i>	<i>15th Norfolk</i>	<i>4/7/2025</i>
<i>Natalie M. Higgins</i>	<i>4th Worcester</i>	<i>4/18/2025</i>
<i>Vanna Howard</i>	<i>17th Middlesex</i>	<i>5/7/2025</i>
<i>Simon Cataldo</i>	<i>14th Middlesex</i>	<i>5/19/2025</i>
<i>Danillo A. Sena</i>	<i>37th Middlesex</i>	<i>5/19/2025</i>
<i>Erika Uyterhoeven</i>	<i>27th Middlesex</i>	<i>5/19/2025</i>
<i>Amy Mah Sangiolo</i>	<i>11th Middlesex</i>	<i>5/21/2025</i>
<i>Michelle L. Ciccolo</i>	<i>15th Middlesex</i>	<i>5/26/2025</i>
<i>Sean Garballey</i>	<i>23rd Middlesex</i>	<i>5/27/2025</i>
<i>Priscila S. Sousa</i>	<i>6th Middlesex</i>	<i>6/6/2025</i>

<i>Kevin G. Honan</i>	<i>17th Suffolk</i>	<i>6/16/2025</i>
<i>David M. Rogers</i>	<i>24th Middlesex</i>	<i>6/16/2025</i>
<i>James B. Eldridge</i>	<i>Middlesex and Worcester</i>	<i>6/16/2025</i>
<i>Margaret R. Scarsdale</i>	<i>1st Middlesex</i>	<i>7/22/2025</i>
<i>John J. Lawn, Jr.</i>	<i>10th Middlesex</i>	<i>12/10/2025</i>

HOUSE No. 3539

By Representatives Owens of Watertown and Armini of Marblehead, a petition (accompanied by bill, House, No. 3539) of Steven Owens, Samantha Montañó and others relative to clean thermal energy. Telecommunications, Utilities and Energy.

The Commonwealth of Massachusetts

In the One Hundred and Ninety-Fourth General Court
(2025-2026)

An Act relative to a tactical transition to affordable, clean thermal energy.

Be it enacted by the Senate and House of Representatives in General Court assembled, and by the authority of the same, as follows:

1 SECTION 1. Section 9 of chapter 25 of the General Laws, as so appearing, is hereby
2 amended by inserting after the word “gas” the following words:-

3 “, utility-scale non-emitting thermal energy”

4 SECTION 2. Section 1 of chapter 164 of the General Laws, as so appearing, is hereby
5 amended by inserting after the definition of “Mitigation” the following definition:-

6 “Non-gas pipe alternative,” a retirement of a gas pipeline, an advanced leak repair of gas
7 infrastructure, or an installation of non-emitting thermal infrastructure, including, but not limited
8 to, electrification of space heating, that: (i) delays, reduces or avoids the need to construct,
9 expand or replace gas infrastructure; and (ii) reduces greenhouse gas emissions as required by
10 chapter 21N.

SECTION 3. Section 94 of said chapter 164 is hereby amended by inserting at the end of the first paragraph the following:-

“The department shall not permit a gas company to recover through rates, prices or charges the costs of construction, expansion or replacement of gas distribution infrastructure unless the gas company demonstrates to the department through its own analysis and through the analysis of requests for proposals solicited through a competitive process from independent entities that all applicable non-gas pipe alternatives were considered and found to be non-viable or cost prohibitive.”

SECTION 4. Said section 94 of said chapter 164 is hereby further amended by striking out, each time they appear, the words “gas or electricity” and inserting in place thereof, in each instance, the following:-

“gas, utility-scale non-emitting thermal energy or electricity”

SECTION 5. Said section 94 of said chapter 164 is hereby further amended by striking out, each time they appear, the words “gas or electric service” and inserting in place thereof, in each instance, the following:-

“gas service, utility-scale non-emitting thermal energy service or electric service”

SECTION 6. Section 106 of said chapter 164 is hereby amended by inserting after the word “chapter” the following:-

“; provided, that the department shall restrict the injection of any amount of a substitute fuel from any source into a gas distribution system that delivers thermal energy to a building unless it determines that such substitute fuel: (i) is non-greenhouse gas emitting in its lifecycle;

(ii) does not pose a safety hazard to persons or property; and (iii) has reliable sources of supply that ensure affordability for customers; and provided further, that the department shall prohibit the injection of any amount of hydrogen into a gas distribution system that delivers thermal energy to a residential, municipal, commercial or other building, except that the department may permit the injection of hydrogen produced from a renewable energy generating source, as defined in section 11F of chapter 25A, into a gas distribution system that delivers thermal energy for an industrial process that is difficult to decarbonize.”

SECTION 7. Section 141 of said chapter 164 is hereby amended by inserting at the end thereof the following:-

(b) In a rate design or other plan for a utility-scale non-emitting thermal energy system filed pursuant to this chapter, the department shall approve a merger of the rate base of such system with the rate base of gas infrastructure and shall permit cross-subsidization between gas ratepayers and utility-scale non-emitting thermal energy system ratepayers.

(c) The department shall not approve a rate design that allows gas companies to recover the costs of line or main extensions to connect new customers to the gas distribution system. A customer requesting new gas service must pay the full capital cost of any such line or main extension.

SECTION 8. Section 145 of said chapter 164 is hereby amended by inserting at the end of subsection (f) the following sentence:-

“The percentage of a gas company’s revenue requirement eligible for recovery that a gas company spends on projects to replace gas infrastructure shall decrease each year by an amount determined by the department until, not later than 2035, 100 percent of the revenue requirement

54 eligible for recovery by a gas company is spent on repair or retirement of gas infrastructure and
55 on installation of utility-scale non-emitting thermal energy projects and 0 percent of the revenue
56 requirement of such company is spent of the replacement of gas infrastructure.”

57 SECTION 9. Said chapter 164, as amended by chapter 239 of the acts of 2024, is hereby
58 amended by inserting at the end thereof the following five sections:-

59 Section 152. (a) As used in this section, the following words shall have the following
60 meanings unless the context clearly requires otherwise:-

61 “Avoided costs”, for a set of buildings that is or will be connected to a utility-scale non-
62 emitting thermal energy system, an estimate of the costs that an electric distribution company
63 would have incurred to instead fully electrify the set of buildings’ energy needs by the current
64 baseline means other than a utility-scale non-emitting thermal energy system.

65 (b) (1) A gas company and an electric distribution company that share service territory
66 shall, not later than October 31 of each year, jointly file with the department a joint tactical
67 thermal transition plan to: (i) reduce greenhouse gas emissions in their shared territory in
68 compliance with the emissions limits and sublimits established in chapter 21N; (ii) promote
69 affordability, including through the maximization of avoided costs and the avoidance of stranded
70 assets; and (iii) promote the additional priorities of the department enumerated in section 1A of
71 chapter 25.

72 (2) A tactical thermal transition plan shall cover the projected work to be performed by
73 street segment over the next 5 years, including, but not limited to: (i) the gas infrastructure to be
74 retired and replaced with utility-scale non-emitting thermal energy systems and the avoided costs
75 resulting from each such installations of a utility-scale non-emitting thermal energy system; (ii)

the gas infrastructure to be retired and replaced by other forms of non-emitting thermal energy, and the cost, for each street segment, to upgrade the electric system to meet the increased need for electricity to serve energy uses formerly served by gas; and (iii) the gas infrastructure to be upgraded or replaced with new gas infrastructure, including, but not limited to, aging or leak-prone gas infrastructure and projects addressing gas constraints, and the associated costs for each street segment. Such a plan shall also include: (i) for each subsequent 5-year period until 2050, the projected number of miles of gas pipe to be retired or replaced and the projected reduction in greenhouse gas emissions achieved during each 5-year period until 2050; (ii) a plan for the hire, retention, and training of an operations and maintenance workforce, in each relevant job classification and department, that is sufficient to fulfill the gas company's obligation to provide safe and reliable gas or utility-scale non-emitting thermal energy service through 2050.

(3) In developing a tactical thermal transition plan, a gas company and an electric distribution company shall solicit recommendations, such as planning scenarios and modeling, from the Thermal Transition Advisory Council established in section 153, respond to information and document requests from such council and conduct technical conferences and a minimum of 2 stakeholder meetings to inform the public, appropriate state and federal agencies and companies engaged in the development and installation of energy efficiency measures, utility-scale non-emitting thermal energy systems, and other building electrification systems.

(c) A gas company and an electric distribution company that share territory shall submit their joint tactical thermal transition plan to the Thermal Transition Advisory Council not later than 150 days before such gas company and electric distribution company file such plan with the department; and provided further, that the Thermal Transition Advisory Council shall return such

plan to such gas company and electric distribution company with recommendations not later than 70 days before the companies file such plan with the department.

A gas company and an electric distribution company that share territory shall submit their joint tactical thermal transition plan, together with a demonstration of the Thermal Transition Advisory Council's review and recommendations, including, but not limited to, a list of each individual recommendation, the status of each recommendation, and an explanation of whether and why each recommendation was adopted, adopted as modified, or rejected, along with a statement of any unresolved issues, to the department in accordance with a schedule determined by the department. The department shall promptly consider such plan and shall provide an opportunity for interested parties to be heard in a public hearing. The department shall approve, approve with modifications, or reject such plan within 60 days of submittal. In order to be approved, a plan shall provide net benefits for customers and meet the criteria enumerated in subsection (b).

(d) The department shall establish and require performance-based rates related to the implementation of the tactical thermal transition plans described in subsections (b). Such performance-based rates shall prioritize the goals of: (i) reducing greenhouse gas emissions in compliance with the limits and sublimits established in chapter 21N; (ii) promoting affordability, including through the maximization of avoided costs and the avoidance of stranded assets; and (iii) promoting the additional priorities of the department enumerated in section 1A of chapter 25. In administering such performance-based rates, the department may deny cost recovery.

(e) Upon approval by the department of the installation of a utility-scale non-emitting thermal energy system by a gas company, the electric distribution company serving the territory

in which the system will be installed shall determine the avoided costs. The electric distribution company shall recover from its ratepayers a portion, determined by the department, of the avoided costs and shall transfer not less than 70 percent of such recovered avoided costs to the relevant program administrator of the energy efficiency program established pursuant to section 21 of chapter 25; provided, that the program administrator shall expend such funds on building retrofits and appliance replacements for customers that are connected to such utility-scale non-emitting thermal energy system; and provided further, that in expending such funds, the program administrator shall give priority to low- and moderate-income customers.

(g) To avoid duplicative infrastructure and stranded assets and to maintain affordability, if a gas company offers utility-scale non-emitting thermal energy to all customers on a street segment, the gas company shall be considered to have met the essential thermal energy service needs of all customers on such street segment.

(h) In any plan or other filing by a gas company that includes the installation of a utility-scale non-emitting thermal energy system or any other type of non-gas pipe alternative, the gas company shall include a plan to provide training and continued employment at pre-existing wages and benefits to workers employed by such gas company whose jobs would otherwise be affected by a transition from gas infrastructure to utility-scale non-emitting thermal energy infrastructure or other non-gas pipe alternatives.

(i) The department shall promulgate such regulations as are necessary to implement this section.

Section 153. (a) There shall be a Thermal Transition Advisory Council to consist of the executive director of the office of energy transformation, or a designee, who shall serve as chair;

the commissioner of energy resources, or a designee; the attorney general, or a designee; the executive director of the Massachusetts clean energy technology center, or a designee; 12 members to be appointed by the governor, 1 of whom shall be a representative of middle-income and low-income residential consumers, 1 of whom shall be a representative from a local agency administering the low-income weatherization assistance program, 1 of whom shall be a representative of the environmental advocacy community, 1 of whom shall be a representative of the thermal transition advocacy community, 1 of whom shall be a representative of an environmental justice community organization, 1 of whom shall be a representative of the energy efficiency industry, 1 of whom shall be a representative of the utility-scale non-emitting thermal energy industry, 1 of whom shall be a representative of the building electrification industry, 1 of whom shall be a representative of the thermal energy workforce, 1 of whom shall be a representative of the gas workforce, 1 of whom shall be a representative of municipal or regional interests, 1 of whom shall have technical and engineering expertise in utility-scale non-emitting thermal energy systems, 1 of whom shall be a representative of businesses, including large commercial and industrial end-use customers, and 1 member from each gas company and electric distribution company operating in the commonwealth, who shall serve as non-voting members. Members shall serve for terms of 5 years and may be reappointed.

(b) The council shall review and provide recommendations on tactical thermal transition plans developed pursuant to section 152. The council shall encourage least-cost investments in the transition of gas distribution customers to sources of non-emitting thermal energy, with the goals of: (i) reducing greenhouse gas emissions in compliance with the limits and sublimits established in chapter 21N; (ii) promoting affordability, including through the maximization of

164 avoided costs and the avoidance of stranded assets; and (iii) promoting the additional priorities of
165 the department enumerated in section 1A of chapter 25.

166 (c) The council shall annually submit to the department a proposal regarding the level of
167 funding required for the retention of expert consultants and reasonable administrative costs. The
168 department shall approve such proposals either as submitted or as modified by the department.
169 The department shall allocate funds sufficient for these purposes from the natural gas and electric
170 energy efficiency funding authorized under section 19 of chapter 25; provided, however, that
171 such allocation shall not exceed 1 per cent of such funding on an annual basis. The consultants
172 retained under this section shall be experts in energy efficiency, utility-scale non-emitting
173 thermal energy, building electrification, or energy finance, and shall be independent.

174 Section 154. (a) A city or town may, by vote of a majority of the city council or by vote
175 of a majority of the select board, establish a clean energy utility: (i) that owns and installs, within
176 the city or town, solar energy systems that qualify as class I net metering facilities, as defined in
177 section 138, and energy storage system with nameplate capacities that are equal to or less than 25
178 kilowatts, on private property with the permission of the private property owner, and that sells
179 the electricity produced or stored by such systems to customers located on the same property as
180 such systems; and (ii) that owns and installs, within the city or town, utility-scale non-emitting
181 thermal energy systems, on public property or on private property with the permission of the
182 private property owner, and sells the thermal energy produced by such systems to customers
183 directly connected to such systems.

184 (b) The department, in consultation with the department of energy resources, shall require
185 an electric distribution company to implement consolidated billing on solar energy systems and

battery storage facilities installed pursuant to subsection (a). In implementing consolidated billing, an electric distribution company shall apply the net value of the bill credit directly to the account of a customer and shall remit directly to the city or town such portion of the payment allocated to such city or town. The net value of the bill credits an electric distribution company applies to an account of a customer may be calculated in a manner determined by the department.

(c) The department shall promulgate such regulations as are necessary to implement this section.

Section 155. To promote and facilitate the safe, efficient and affordable development of utility-scale non-emitting thermal energy systems throughout the commonwealth and to ensure that thermal energy is managed for the public trust, the department shall promulgate regulations governing the installation, construction, operation and maintenance of utility-scale non-emitting thermal energy systems, except those that are installed and operated wholly on private property.

Section 156. (a) As used in this section, the following words shall have the following meanings unless the context clearly requires otherwise:-

“Electrification project”, a method of providing a set of gas customers with suitable non-emitting thermal energy that does not involve utility-scale infrastructure. Such a method may include, but shall not be limited to, the replacement of gas appliances with electric appliances, energy efficiency measures, and demand flexibility measures to alter energy consumption.

(b) (1) On or before January 1, 2027, the department shall designate priority neighborhood electrification zones. In designating the such zones, the department shall consider factors that include, but are not limited to, the following: (i) the tactical thermal transition plans

approved pursuant to section 152; (ii) the availability of supportive municipal government or community partners; (iii) the concentration of low- and moderate-income customers; and (iv) the concentration of projects to replace gas distribution pipes pursuant to section 145.

(2) The department shall coordinate with relevant agencies to identify non-ratepayer funding, such as state and federal funds, that may be used to implement electrification projects in priority neighborhood electrification zones.

(3) The department may direct a gas company and an electric distribution company, if appropriate, to obtain resources for an electrification project from other available programs, including, but not limited to, energy efficiency, low-income weatherization, gas system enhancement, and distributed generation.

(4) The department may update the priority neighborhood electrification zones as necessary.

(c) (1) On or before January 1, 2027, in a new or existing proceeding, the department, in consultation with gas companies and electric distribution companies, shall establish a program to facilitate the cost-effective decarbonization of priority neighborhood electrification zones through electrification projects.

(2) In administering such projects, the department shall establish all of the following: (i) a process for a gas company to determine and submit electrification projects for approval by the department; (ii) criteria and methodology for determining the cost-effectiveness of electrification projects as compared to replacement, repair, or continued operation of the affected asset of the gas system; provided, that nonenergy benefits may be considered; and provided further, that the cost incurred by a gas company for the electrification project shall be less than the cost that

would have been incurred had a gas pipe been replaced; (iii) requirements and programs to ensure that a substitute for gas service for low-income customers is affordable, adequate, efficient, and just and reasonable; (iv) a preference for projects that provide prevailing wages; (v) a requirement that a gas company recover costs related to a project that are deemed just and reasonable and a requirement that prohibits a gas company from recovering behind-the-meter costs associated with a project as capital costs that are afforded a rate of return; and (vi) the appropriate rate of return and recovery period that a gas company is eligible to receive for its costs to implement an electrification project; provided, that a gas company shall not receive ratepayer funding for the costs of an electrification project that are covered by incentives under federal, state, or local laws.

(3) To avoid duplicative infrastructure and stranded assets and to maintain affordability, if a gas company implements an electrification project pursuant to subsection (c) and offers suitable substitute energy service to all customers in the priority neighborhood electrification zone, the gas company shall be considered to have met the essential thermal energy service needs of all customers within such priority neighborhood electrification zone.

(d) (1) Beginning on January 1, 2030, the department, in a new or existing proceeding, shall review the efficacy of the electrification projects established pursuant to subsection (c) in providing benefits to customers of a gas company and in assisting the commonwealth in achieving compliance with the statewide greenhouse gas emission limits and sublimits established under chapter 21N. On or before March 1, 2031, the department shall submit to the relevant committees of the general court a report on the review.

251 (2) On or before March 1, 2027, and on or before March 1 of each year thereafter, the
252 department shall submit a progress report to the relevant committees of the general court
253 summarizing the findings of the electrification projects, including the locations of the projects,
254 the number of customers affected, the costs of the projects, the funding used to pay for the
255 projects, any assistance provided to customers, and any outcomes, challenges, and
256 recommendations.